

SOLUTION REQUIREMENTS

1. Project Overview

The AI FAQ Assistant is a web-based academic application designed to help users obtain relevant answers from a centralized FAQ knowledge base. The system combines user authentication, category-based FAQ organization, administrative management, and an AI-assisted text matching mechanism.

2. Project Team

Team ID	SWTID-2026-9997
Team Size	5
Team Leader	Shalini K
Team Member	Sindhuja S
Team Member	Sabrin Nisha K
Team Member	Yamini A
Team Member	Sankari N

3. Functional Requirements

FR-01	User Registration	The system shall allow a new user to register with name, email and password.
FR-02	User Login	The system shall authenticate registered users and provide a secure session using JWT-based authentication.
FR-03	User Profile	Authenticated users shall be able to retrieve their current account information.
FR-04	View FAQs	Users shall be able to retrieve available FAQs from the system.
FR-05	View Categories	Users shall be able to retrieve FAQ categories.
FR-06	Ask AI Assistant	Users shall be able to submit a question to the AI Assistant.
FR-07	FAQ Matching	The system shall compare the question with FAQ questions, keywords and answers and identify relevant matches.
FR-08	Related Suggestions	The AI response may provide related FAQ suggestions based on matching results.
FR-09	Admin FAQ Management	Authorized administrators shall be able to create, update and delete FAQ entries.
FR-10	Category Management	Authorized administrators shall be able to create, update and delete categories.
FR-11	User Management	Authorized administrators shall be able to view users, view user statistics, delete users and update user roles.
FR-12	FAQ View Tracking	FAQ records shall maintain a view count for tracking usage.

4. Non-Functional Requirements

Performance	The application should process FAQ requests and matching operations efficiently.
Security	Passwords must be protected and administrative operations must require authentication and authorization.
Usability	The interface should provide simple navigation for students and administrators.
Maintainability	The application should use modular controllers, routes, models and utilities.
Scalability	The architecture should allow additional FAQs, categories and users to be added.
Reliability	Errors should be handled through centralized middleware and meaningful API responses.
Data Integrity	FAQ, category and user data should be stored consistently in MongoDB.

5. Business Rules

- User roles are **user** and **admin**.
- FAQ and category modifications require administrative authorization.
- User passwords are not returned in normal user objects.
- The AI matching engine returns a relevant FAQ only when its matching score reaches the configured threshold.
- FAQ records are associated with categories.

6. Requirement Summary

The requirements establish a secure, role-based FAQ platform in which students can search and ask questions while administrators maintain the underlying FAQ knowledge base.